A Gaussian meter breaks the classical entropy-floor witness

Result, 2026-10-28; written derivation, unrefereed. The classical entropy-floor class used as a witness in the canonical-averaging note is not closed under a single selective Gaussian position measurement. The initial body and pointer both obey the same positive entropy and covariance-action floor. The pointer has independent Gaussian error and kick with the minimum action product $S_0\zeta$, and the body–pointer interaction is a reversible canonical shear. A single record event has probability at least 9/32, yet its body posterior can have covariance action arbitrarily close to zero. Both unread one-pair marginals remain in the entropy-floor class.

For a Gaussian body preparation the same meter *does* preserve the floor, even after conditioning on any measurable record event. This exact contrast identifies the hidden issue: the entropy floor admits non-Gaussian mixtures containing a very narrow component, concealed by a broad component. With $\hbar = 2S_0\zeta$, those mixtures are not Wigner functions of quantum states, although they are positive, have finite covariance and satisfy the same global entropy and covariance inequalities. A coherent-state overlap proves their nonquantumness below.

A narrower classical class—the W_2 closure of finite mixtures of admissible Gaussians—survives this meter, affine canonical changes and copy-index mixers. It retains the supplied floor, but a smooth nonlinear canonical shear takes one of its Gaussian states outside the class. A three-point kernel test proves the failure without invoking a numerical calculation or a general no-go theorem.

The construction tests one specified classical instrument, not all physically possible readouts. Its initial body aperture becomes unbounded along the zero-action sequence. The entropy threshold is supplied, and no positive $\hbar$ is derived. The underlying Gaussian equivalence has substantial prior art: Bartlett, Rudolph and Spekkens 2012 (publisher abstract read) recover Gaussian quantum statistics from a classical uncertainty condition plus maximum entropy at fixed moments. The two-pointer theorem already defeats a prior density ceiling under explicit instrument-access assumptions. Here the target is the later entropy witness and one meter; no literature priority is claimed.

1. Entropy floor and a canonical meter

Choose calibration units $L_0, P_0 > 0$, $S_0 = L_0P_0$, and use dimensionless canonical variables $Q = q/L_0$, $P = p/P_0$ and similarly Y, Π for a separate pointer. A covariance matrix C of (Q, P) has physical covariance action $S_0\sqrt{\det C}$. Fix a **supplied** $\zeta > 0$ and write

$$G_v(z) = \frac{1}{2\pi v}\exp\left(-\frac{|z|^2}{2v}\right), \qquad \mathcal{E}_\zeta = \left\{\rho\, dz \in \mathcal{P}_2(\mathbb{R}^2) : h(\rho) \geq \log(2\pi e\zeta)\right\}, \tag{1}$$

where $h(\rho) = -\int \rho \log \rho\, dz$ is dimensionless differential entropy. The Gaussian maximum-entropy inequality implies

$$\rho \in \mathcal{E}_\zeta \quad \Longrightarrow \quad S_0\sqrt{\det \mathrm{Cov}_\rho(Q, P)} \geq S_0\zeta. \tag{2}$$

Prepare the pointer independently with density $G_\zeta(Y, \Pi)$. Thus its error Y and kick Π have variances ζ and their physical standard-deviation product is $S_0\zeta$. The canonical interaction is

$$Q' = Q, \quad P' = P - \Pi, \quad R = Y' = Y + Q, \quad \Pi' = \Pi. \tag{3}$$

It is the unit-strength Hamiltonian pulse generated by $S_0Q\Pi$: the common dimensionless symplectic form is $S_0(dQ \wedge dP + dY \wedge d\Pi)$. Equation (3) is invertible and preserves its four-dimensional phase volume. We record the pointer coordinate R, retaining either its value or an event A of positive probability. The instruments and independent pointer preparation are explicit availability assumptions.

2. Exact Gaussian record closure

Theorem 1. Let the body input be a Gaussian with covariance $C = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$, where $a > 0$ and $\det C \geq \zeta^2$. Under (3), every exact pointer outcome has a Gaussian body posterior with covariance

$$C_{\mathrm{post}} = \begin{pmatrix} a\zeta/(a+\zeta) & c\zeta/(a+\zeta) \\ c\zeta/(a+\zeta) & b+\zeta-c^2/(a+\zeta) \end{pmatrix}, \quad \det C_{\mathrm{post}} = \zeta \frac{\det C + a\zeta}{a+\zeta} \geq \zeta^2. \tag{4}$$

For every Borel record event A with positive probability, the postselected body law also belongs to $\mathcal{E}_\zeta$.

Proof. The output body covariance before reading is $\begin{pmatrix} a & c \\ c & b+\zeta \end{pmatrix}$, its covariance with R is $(a, c)^{\mathsf{T}}$, and $\operatorname{Var} R = a + \zeta$. Gaussian conditioning gives (4). Its last inequality is equivalent to $\det C \geq \zeta^2$. The conditional covariance does not depend on the outcome. Conditioning instead on A mixes those Gaussian conditional laws, all with the same covariance and Gaussian entropy $\log(2\pi e\sqrt{\det C_{\mathrm{post}}})$. Entropy concavity gives the stated floor for the mixture. □

Thus this meter is a classical realization of the quantum Gaussian update on the Gaussian sector. Closure for that sector does not settle closure for the larger class (1).

3. A non-Gaussian preparation selected by one ordinary record

For $0 < \delta \leq 1/2$ put

$$s = \zeta\delta, \qquad b_\delta = \zeta\delta^{-3}, \qquad d = \sqrt{\zeta}\,\delta^{-2}, \qquad \rho_\delta = \tfrac{3}{4}G_s(Q,P) + \tfrac{1}{4}G_{b_\delta}(Q-d,P). \tag{5}$$

All variances are finite for each $\delta > 0$. By concavity of entropy,

$$h(\rho_\delta) \geq \tfrac{3}{4}\log(2\pi e s) + \tfrac{1}{4}\log(2\pi e b_\delta) = \log(2\pi e\zeta). \tag{6}$$

The input body and pointer therefore both belong to $\mathcal{E}_\zeta$. Take the finite-resolution record event

$$A = \{|R| \leq \sqrt{\zeta}\}. \tag{7}$$

Theorem 2. Let w_δ be the posterior weight of the broad component in (5) after event (7), and let κ_δ be the physical covariance action of the postselected body. Then

$$\mathbb{P}(A) \geq \frac{9}{32}, \qquad 0 \leq w_\delta \leq \delta^{3/2} e^{-1/(4\delta)}, \tag{8}$$

and

$$\frac{\kappa_\delta}{S_0\zeta} \leq \sqrt{(\delta + \delta^2 + 5w_\delta)(1 + \delta + w_\delta\delta^{-3})} \;\longrightarrow\; 0. \tag{9}$$

Consequently the posterior lies outside $\mathcal{E}_\zeta$ for all sufficiently small δ, despite the fixed positive pointer floor.

Proof. In the narrow and broad components respectively,

$$R_s \sim N(0, \zeta(1+\delta)), \qquad R_b \sim N(d, \zeta(1+\delta^{-3})). \tag{10}$$

Let p_s, p_b be their probabilities of A. Since $1/\sqrt{1+\delta} \geq 1/\sqrt{2}$, the normal integral gives

$$p_s \geq \frac{e^{-1/4}}{\sqrt{\pi}} \geq \frac{3}{8}. \tag{11}$$

For $0 < \delta \le 1/2$, the broad normal density on the interval is at most its value at $\sqrt{\zeta}$, so

$$p_b \le \sqrt{\frac{2}{\pi}}\,(1+\delta^{-3})^{-1/2} \exp\left[-\frac{(\delta^{-2}-1)^2}{2(1+\delta^{-3})}\right] \le \delta^{3/2} e^{-1/(4\delta)}. \tag{12}$$

The second inequality uses $(1-\delta^2)^2/[2\delta(1+\delta^3)] \ge 1/(4\delta)$. The total event probability is $(3p_s + p_b)/4 \ge 9/32$, and $w_\delta = p_b/(3p_s+p_b) \le p_b$, proving (8).

For the narrow component, Gaussian conditioning on $R = r$ gives conditional mean $\delta r/(1+\delta)$ and variance $\zeta\delta/(1+\delta)$. Averaging over $|r| \le \sqrt{\zeta}$ yields

$$\mathbb{E}(Q^2 \mid A, \text{narrow}) \le \zeta(\delta + \delta^2). \tag{13}$$

For the broad component the conditional variance is $\zeta/(1+\delta^3)$, and the conditional mean is $[\sqrt{\zeta}\,\delta + r]/(1+\delta^3)$. On A its absolute value is at most $2\sqrt{\zeta}$, so the corresponding second moment is at most 5ζ. The final momentum $P' = P - \Pi$ is, within either component, independent of (Q, R), centered, and has variance $s + \zeta$ or $b_\delta + \zeta$. Hence

$$\mathrm{Var}(Q \mid A) \le \zeta(\delta + \delta^2 + 5w_\delta), \quad \mathrm{Var}(P' \mid A) \le \zeta(1 + \delta + w_\delta \delta^{-3}). \tag{14}$$

The covariance determinant is no larger than the product of these variances, which proves (9). Since $w_\delta \delta^{-3} \to 0$, the bound tends to zero. Equation (2) excludes the posterior from the entropy-floor class. □

The failure is selective. Before reading R, the body law is the input law averaged over independent momentum translations, so entropy concavity gives $h(Q, P-\Pi) \ge h(Q, P)$. The pointer's unread law has $R = Q + Y$ independent of Π, and $h(R) \ge h(Y)$ by the same argument. Both unread marginals therefore remain in $\mathcal{E}_\zeta$; the full joint entropy is conserved by the canonical shear. Selective closure is a strictly stronger requirement than this unread evolution.

The counterexample uses no divergent quantity at any fixed δ. Its broad-component variance and position separation grow as $\zeta\delta^{-3}$ and $\sqrt{\zeta}\,\delta^{-2}$, respectively. Thus it does not defeat a separately imposed finite preparation aperture or energy budget uniformly in δ; the resource-relative distinction remains.

4. Why the positive classical density is not a quantum state

Set the *physical* reduced action to $\hbar = 2S_0\zeta > 0$, so the commutator of the calibrated variables is $[\widehat{Q}, \widehat{P}] = i\hbar/S_0 = 2i\zeta$. Let σ_0 be the pure minimum-uncertainty Gaussian state whose normalized Wigner function is G_ζ. If (5) were the Wigner function of a density operator ρ, the Wigner trace identity would give

$$\mathrm{Tr}(\rho\sigma_0) = 2\pi\frac{\hbar}{S_0}\int \rho_\delta(z) G_\zeta(z)\,dz \ge \frac{3}{2(1+\delta)} > 1, \qquad 0 < \delta < \tfrac{1}{2}. \tag{15}$$

Here $\int G_s G_\zeta\,dz = [2\pi(s+\zeta)]^{-1}$, and the broad component contributes a nonnegative amount. Equation (15) is impossible because a density operator has expectation at most one in a normalized pure state. The overlap identity is the usual Wigner Hilbert–Schmidt formula; see Bartlett–Rudolph–Spekkens, §IV.B, equation (90) (passage read; the factor in (15) uses our unit-integral Wigner convention). This is an explicit positivity test beyond the entropy and covariance inequalities, not a new characterization of all Wigner functions.

Under the quantum identification above, the meter G_ζ itself is a legitimate minimum-uncertainty state and (3) is the Wigner action of the quadratic unitary pulse. The counterexample cannot be imported as a quantum preparation: its concentrated component makes the full input operator nonpositive. In particular, positivity of a classical phase-space density, a covariance floor, entropy closure and Gaussian-meter back-action do not alone supply quantum state positivity.

5. A record-stable classical repair and its nonlinear obstruction

Let $\mathcal{C}_\zeta$ be the finite convex mixtures of one-pair Gaussian probabilities with $\det C \geq \zeta^2$, and let $\overline{\mathcal{C}}_\zeta$ be its W_2 closure in $\mathcal{P}_2$.

Theorem 3. The class $\overline{\mathcal{C}}_\zeta$ has physical covariance-action infimum $S_0\zeta$ and is closed under affine canonical transformations, independent-copy orthogonal canonical mixing followed by retention of one pair, and the Gaussian meter (3) followed by any positive-probability Borel record event. Nevertheless, for every $\lambda \neq 0$ the classical canonical shear

$$Q' = Q, \qquad P' = P + \lambda Q^2 \tag{16}$$

sends $G_\zeta \in \overline{\mathcal{C}}_\zeta$ to a density outside that class, although the output remains in $\mathcal{E}_\zeta$.

Proof of the closed operations. Entropy concavity and (2) put every finite Gaussian mixture in $\mathcal{E}_\zeta$, and the latter is W_2-closed by the central-limit note, §6. Thus the closure has action at least $S_0\zeta$; G_ζ attains it.

An affine canonical map carries every admissible Gaussian to another. For one output of an orthogonal copy-index mixer, the covariance is $\sum_j a_j^2 C_j$ with $\sum_j a_j^2 = 1$. Each $C_j + i\zeta J \succeq 0$, $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so their convex combination also obeys that inequality, equivalent in one pair to $\det C \geq \zeta^2$. Finite input mixtures therefore give finite output mixtures. These operations are continuous in W_2 on independent product laws, so they extend to the closure.

For a finite Gaussian mixture, Theorem 1 shows that conditioning on a record value gives a finite mixture of admissible Gaussians. Conditioning on a Borel event A gives an integral mixture over the record values. Its record distribution has finite second moment; approximation by finitely supported distributions makes this an element of the W_2 closure. To extend the result to all input laws in the closure, write the event likelihood as

$$\ell_A(q) = \int_A \frac{e^{-(r-q)^2/(2\zeta)}}{\sqrt{2\pi\zeta}}\, dr. \tag{17}$$

If A has positive probability, it has positive Lebesgue measure; ℓ_A is bounded, continuous and strictly positive. The posterior is the input law weighted by $\ell_A(Q)$, normalized, followed by convolution of P with the independent Gaussian kick. W_2 convergence of inputs implies weak convergence plus uniformly integrable second moments, so these weighted normalized laws converge in W_2; coupling the same kick preserves that convergence. Exact record values use the bounded positive Gaussian likelihood at that value. Hence all the claimed record outputs remain in $\overline{\mathcal{C}}_\zeta$.

Proof of the nonlinear obstruction. Equation (16) is a global canonical map, generated by the pulse $-S_0\lambda Q^3/3$. Its output is

$$f_\lambda(q,p) = \frac{1}{2\pi\zeta} \exp\left[-\frac{q^2 + (p - \lambda q^2)^2}{2\zeta}\right]. \tag{18}$$

The shear preserves entropy, so $h(f_\lambda) = \log(2\pi e\zeta)$; it has finite covariance and belongs to $\mathcal{E}_\zeta$.

Every member of $\mathcal{C}_\zeta$ is a Wigner function of a positive quantum state when $\hbar = 2S_0\zeta$. For every Schwartz test state ψ, its Wigner function is bounded and continuous, and the Wigner overlap with any member of $\mathcal{C}_\zeta$ is nonnegative. This necessary condition is preserved by weak convergence, hence by W_2 closure. We now violate it for (18).

The inverse Wigner kernel corresponding to (18) is continuous and equals

$$K_\lambda(x,y) = \frac{e^{-(x^2+y^2)/(4\zeta)}}{\sqrt{2\pi\zeta}} \exp\left[\frac{i\lambda(x+y)^2(x-y)}{8\zeta}\right]. \tag{19}$$

If it came from a positive operator, every finite matrix $[K_\lambda(x_i, x_j)]$ would be positive semidefinite; smooth test bumps approximating point masses justify this necessary condition. Normalize the diagonal of the three-point matrix at $x_1 = -a, x_2 = 0, x_3 = a$. The normalized off-diagonal entries have modulus one and cyclic phase $-\lambda a^3/(4\zeta)$, so its determinant is

$$2\left[\cos\left(\frac{\lambda a^3}{4\zeta}\right) - 1\right]. \tag{20}$$

Choosing $a = (4\pi\zeta/|\lambda|)^{1/3}$ makes (20) equal -4. A finite combination of smooth compactly supported bumps therefore has negative expectation against K_λ. Its Wigner function is a bounded continuous test violating the necessary condition above, so $f_\lambda \notin \overline{\mathcal{C}}_\zeta$. □

This is a concrete boundary between the two classical repairs: the larger $\mathcal{E}_\zeta$ survives this nonlinear Liouville shear but fails selective Gaussian recording; the smaller $\overline{\mathcal{C}}_\zeta$ survives that record but not the shear. Neither statement excludes every other classical completion. In particular ζ is supplied in both constructions.

The distinction also has an exact quantum form. The quantum pulse for $-S_0\lambda Q^3/3$ sends the minimum Gaussian wavefunction to $\psi(q)\exp[i\lambda q^3/(6\zeta)]$. Its density kernel is

$$K_{\text{quantum}}(x, y) = K_\lambda(x, y)\exp\left[\frac{i\lambda(x-y)^3}{24\zeta}\right]. \tag{21}$$

The additional off-diagonal phase is absent in the purely classical shear. Equation (21) is a direct calculation at fixed physical $\hbar = 2S_0\zeta$. On separations $|x - y| = O(\sqrt{\zeta})$ the phase is $O(|\lambda|\sqrt{\zeta})$ as $\hbar/S_0 = 2\zeta \to 0$. The three-point test at fixed λ instead uses

$$\frac{a}{\sqrt{\zeta}} = \left(\frac{4\pi}{|\lambda|}\right)^{1/3}\zeta^{-1/6}, \qquad \frac{K_\lambda(a, a)}{K_\lambda(0, 0)} = \exp\left[-\frac{1}{2}\left(\frac{4\pi}{|\lambda|}\right)^{2/3}\zeta^{-1/3}\right]. \tag{22}$$

Thus the normalized determinant stays at -4 while the test points move into suppressed Gaussian tails. It gives no lower bound on an operational distance that remains finite in the large-action limit. As in the earlier cusp comparison, S_0 must describe a physical family of characteristic actions for $\hbar/S_0 \to 0$ to mean more than a change of units.

6. Consequence for STATE

The entropy-floor witness satisfies the existing copy and actual-cusp transport tests, but a single allowed Gaussian meter and a positive- probability coarse record break its selective closure. The Gaussian-mixture repair survives the record and copy tests but fails a nonlinear classical canonical shear. Quantum operator positivity blocks both exhibited nonphysical densities, with $\hbar$ supplied; quantum nonlinear evolution also differs from the classical shear by (21). Independently deriving a state and dynamics law, its physical scale and closure for actual records remains open.